LAHARZ Olympics

Homework- Day One

1. With your team, pick a volcano where you want to make a lahar hazard map.
2. Review the eruption history of your volcano using the Smithsonian Institution GVP website (http://www.volcano.si.edu/) and/or observatory reports.
   - What is typical behavior for an eruption in recent history (VEI, domes/flows/magma type/effusive/explosive)?
   - What are the potential larger, longer-term styles of eruption?
3. Gather information about lahars at your volcano:
   - lahar history, volumes and recurrence frequency (how big? how often?)
   - what are the triggers for lahars (typhoon, rainstorm, earthquake, eruption, landslide, pyroclastic flow, rapid snow melt, dome collapse, avalanche)
   - where have lahars been mapped or reported?
4. Is there already a lahar hazard map?
   - Is there perhaps a good reason to make a new map for a different scenario or different audience?
   - Use https://volcanichazardmaps.org/ to look at maps for your volcano and other maps you might like
5. Summarize potential lahar hazards based on this historic information. Discuss what scenario you would like to portray.

Homework- Day Two

1. Review any spatial information about your volcano.
    - geologic map (where are the pyroclastic flows, lava flows, river channels, areas for downstream flooding)
    - topographic map (where are the steep areas or steep canyons- where will a flow be well contained and where will a flow perhaps spill over into a flat area?) what does the geomorphology tell you?
    - Use Google Earth to explore your volcano.
    - Make an elevation profile and think about the H/L value you might use
2. Where are the people?
    - Find a population map. How many people live nearby?
3. Identify major roads, rivers, lakes, villages, major cities. Where are the most important, critical places if a disaster happens?
    - Airports, energy resources, military assets, hospitals, schools and evacuation centers, sensitive natural areas
4. Decide on the scenario or scenarios for hazard map:
    - What type of lahars are you portraying (rain-fall triggered, eruption triggered, etc?)
    - If eruption-triggered, what size eruption are you considering?
5. Decide on your target audience:
    - General public? Mining company or transportation agency? Tourists? Emergency managers? Land-use planners?

Judging Criteria

- Map should be beautiful and easy to understand.
- Put your names on it. Stake your professional reputation on it.
- Scale bar, north arrow, location map or spatial context (latitude, longitude)
- Metadata on DEM (source, resolution, projection), cite references or make acknowledgements to other data sources
- Should be easy to recognize scenario and your message
- Should be clear who your audience is: technical or non-technical?
- Appropriate colors with good design elements.
- Logos for partner agencies; make it authoritative.
- Areas of relative hazard easy to understand (high, medium, low)
- Show some indication of your level of uncertainty and possible error
- Oblique photo and/or 3-D view are optional but helpful
- Be creative and have fun!